GRINNELL DRY PIPE VALVE ACCELERATOR

MODEL A-6

FOR USE WITH MODEL F302 & F3021 DRY PIPE VALVES

Instructions
for
Installation
Care and Maintenance

OBSOLETE

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DRY PIPE VALVE ACCELERATOR • MODEL A-6

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DESCRIPTION

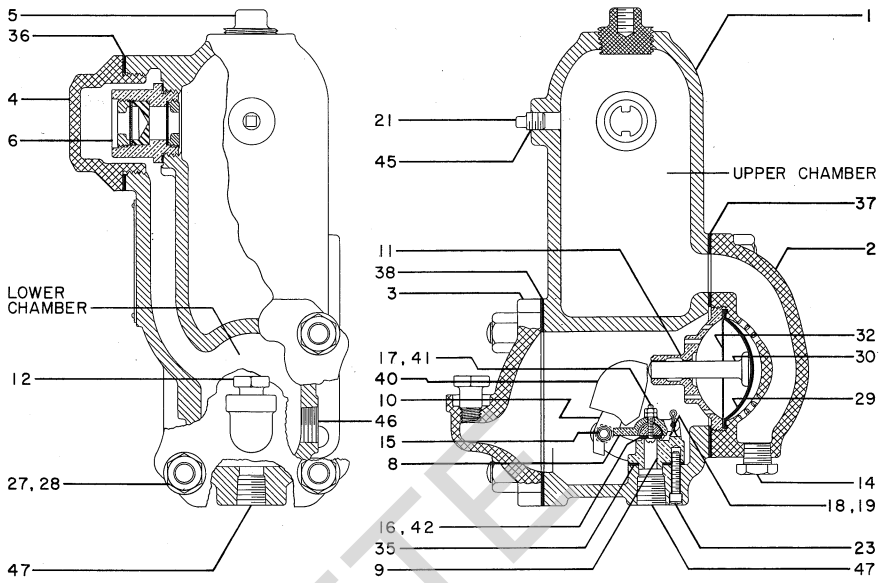
The Model A–6 Dry Pipe Valve Accelerator is a pressure sensitive quick opening device used in dry pipe sprinkler systems with capacities of more than 500 gallons. When system capacities exceed 600 gallons, an Extra Chamber connected to the Accelerator is required to retain its sensitivity to pressure.

Rated for 175 psi pressure service, the Accelerator is designed for use with differential type dry pipe valves to accelerate operation of the valves by destroying the air to water differential which normally holds them closed.

Trimmings for the Accelerator include an Anti-Flooding Device to prevent contamination of the Accelerator with debris from the system when it operates.

APPROVALS

The Model A–6 Accelerator is listed by Underwriters Laboratories, Inc. and Underwriters' Laboratories of Canada. It is approved by Factory Mutual and the Fire Offices Committee.



1—Body
2—Diaphragm Plate
3—Handhole Plate
4—Restriction Chamber Cap
5—1¼" Plug
6—Restriction Chamber Assembly
8—Valve Cage
9—Valve Body
10—Weight Lever
11—Diaphragm Rod Guide
12—Priming Plug
14—Air Chamber Plug
15—Weight Lever Screw (2 req'd)
16—Valve Washer
17—Machine Screw, 8–32 UNC x 1"
18—Valve Pin
19—Cotter Pin, 1/16" x 3/8" (2 req'd)
21—¼" Plug
23—Fillet Head Machine Screw, ¼"–20 UNC x 1¼" (3 req'd)
27—Stud Bolts, 5/8"–11 UNC x 2 3/8" (10 req'd)
28—Hex Nuts, 5/8"–11 UNC (10 req'd)
29—Diaphragm
30—Diaphragm Rod
32—Diaphragm Washer
35—Valve Gasket
36—Restriction Chamber Cap Gasket
37—Diaphragm Plate Gasket
38—Handhole Plate Gasket
40—Weight
41—Hex Nuts, 8–32 UNC (2 req'd)
42—Valve Plug
45—Air Pressure Gauge Connection, ¼" NPT
46—System Air Inlet Connection, ¾" NPT
47—Dry Pipe Valve Connection, ½" NPT

FIGURE A
MODEL A–6 ACCELERATOR

DESIGN (Fig. A)

Principal components of the Accelerator are the Restriction Chamber, Valve, Diaphragm Plate and Body Assemblies. Although not specifically components of the Accelerator, Extra Chambers and an Anti-Flooding Device are accessories which significantly contribute to its efficient operation.

Restriction Chamber Assembly. The Restriction Chamber Assembly (6), mounted in the upper chamber of the Body (1) of the Accelerator, consists of a Restriction Chamber Orifice, two Screens and Gaskets mounted in the Restriction Chamber Body with two Restriction Chamber Nuts.

Valve Assembly. The Valve Assembly consists of a Valve Closure (8, 16, 17, 41, 42) mounted to the Valve Body (9) with a Valve Pin (18) and two Cotter Pins (19), and a Weight Assembly (10, 40) mounted to the Valve Body with two Weight Lever Screws (15). The Valve Assembly is mounted in the lower Chamber with a Gasket (35) and three Fillet Head Screws (23).

Diaphragm Plate Assembly. The Diaphragm Plate Assembly consists of a Diaphragm (29), Diaphragm Washer (32), Diaphragm Rod (30) and a Diaphragm Rod Guide (11) mounted to the Diaphragm Plate (2). An Air Chamber Plug (14) is screwed into the lower section of the Diaphragm Plate.

Body Assembly. The Body Assembly consists of the Body (1), Restriction Chamber Cap (4), Diaphragm Plate (2) and the Handhole Plate (3) in which there is a Priming Plug (12). The Restriction Chamber Cap, with its Gasket (36), screws into the upper part of the Body. The Diaphragm and Handhole Plates, with their Gaskets (37, 38) are mounted to the lower part of the Body with Stud Bolts (27) and Hex Nuts (28).

The Body is divided into two chambers separated by the Restriction Chamber Assembly (6) and the Diaphragm Plate Assembly. Communication between the chambers is through the orifice in the Restriction Chamber Assembly.

There are three female NPT Connections (45, 46, 47) in the Body for an Air Pressure Gauge and for attachment to the Accelerator to the dry pipe valve and to the system piping.

CAPACITY IN GALLONS			DIM. IN INCHES	
EXTRA CHAMBER		DRY PIPE SYSTEM	LENGTH OF	A
NO.	CAPACITY	CAPACITY	4" PIPE	
2	.66	600- 799	12	14
4	1.05	800-1074	19	21
6	1.63	1075-1399	29 ½	31 ½
8	2.07	1400-1600	37 ½	39 ½

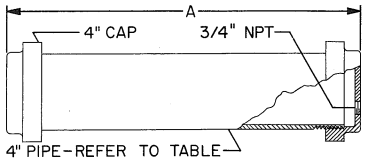
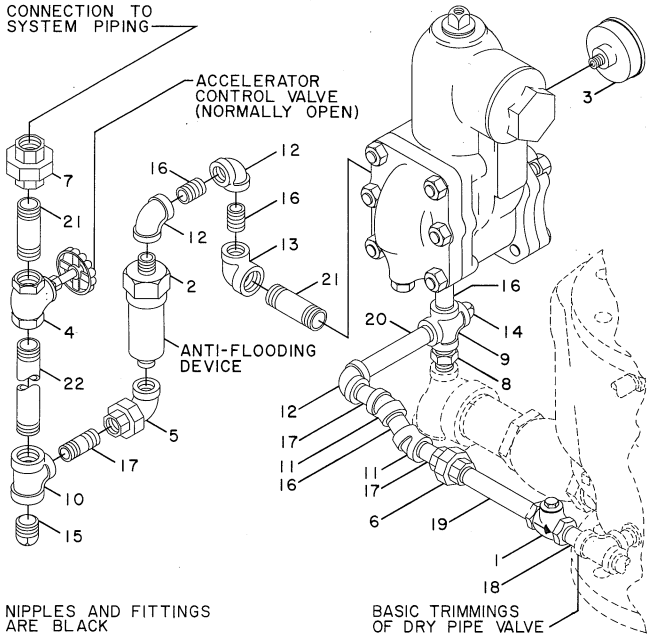


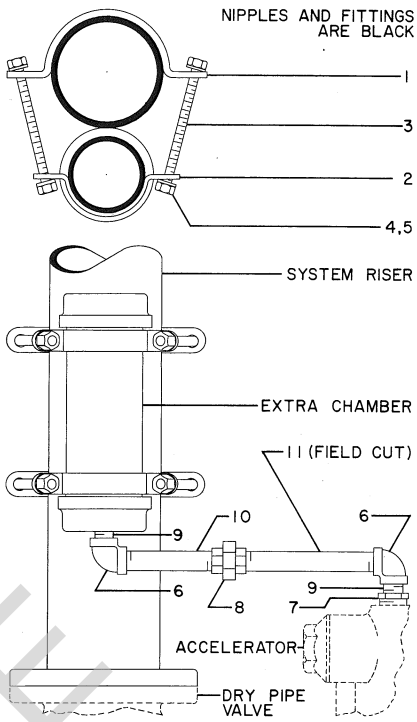
FIGURE B
EXTRA CHAMBER

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1—½" Check Valve, United Brass, Fig. 99S
2—Accelerator Anti-Flooding Device, Standard Model A
3—Accelerator Air Pressure Gauge
4—¾" Globe Valve, United Brass, 125S
5—½" 90° Union Elbow, Fig. 552
6—½" Union, Fig. 463
7—¾" Union, Fig. 463
8—Accelerator Support Plug
9—½" Cross
10—¾" x ¾" x ½" Tee
11—½" 45° Elbow (2 req'd)
12—½" 90° Elbow (3 req'd)
13—¾" x ½" 90° Reducing Elbow
14—½" Square Head Plug
15—¾" Square Head Plug
16—½" x close Nipple (4 req'd)
17—½" x 2" Nipple (3 req'd)
18—½" x 2" Nipple for 4" valves, ½" x 3" Nipple for 6" valves
19—½" x 4½" Nipple
20—½" x 5" Nipple
21—¾" x 3" Nipple (2 req'd)
22—¾" x 10" Nipple

FIGURE C
BASIC TRIMMINGS FOR MODEL A–6 ACCELERATOR



1—Pipe Clamp, Fig. 212 (same size as system riser)
2—4" Pipe Clamp, Fig. 212
3—Machine Thread Rod, (4 req'd) ½" x 8" for 4" riser, ½" x 9" for 5" and 6" risers, ½" x 10" for 8" riser
4—½" Hex Nuts (8 req'd)
5—½" Standard Washers (8 req'd)
6—¾" 90° Elbows (2 req'd)
7—1¼" x ¾" Hex Bushing
8—¾" Union, Fig. 463
9—¾" x 1½" Nipple (2 req'd)
10—¾" x 8" Nipple
11—¾" x 9" Nipple, T.O.E.

FIGURE D
EXTRA CHAMBER TRIMMINGS

An Identification Label (not shown) with the Accelerator model and serial number is mounted on the Body. Other identifications are cast in raised print on the Body and Diaphragm Plate. All exterior surfaces, except the Identification Label, are painted red.

Extra Chamber. Extra Chambers consist of a length of pipe and two caps assembled as shown in Figure B. Identification numbers, capacities and dimensions of Extra Chambers are listed in the table in Figure B.

Anti-Flooding Device. The Anti-Flooding Device (2—Fig. C) is designed to remain open to permit a free flow of pressure between the system piping and the Accelerator when the system is set for service, and to close upon system operation to prevent the flow of foreign material to the Accelerator.

INSTALLATION

Figure C shows the bill of materials and the configuration of the trimmings for the Accelerator when required for installation with the Model F302/F3021 Dry Pipe Valve.

The Accelerator must be mounted plumb and the connection to the system piping must be at least two feet above the level of the priming water in the dry pipe valve.

If an Extra Chamber is required, mount the Chamber to the system riser, remove the Plug (5—Fig. A) from the top of the Accelerator and connect the Chamber to the Accelerator as shown in Figure D.

If installation conditions prevent mounting the Extra Chamber as shown in Figure D, it may be mounted in a vertical or horizontal position to an adjacent wall or other suitable mounting surface. It must not be mounted to a surface subject to vibration and the connections to the Accelerator must be arranged to permit drainage from the Extra Chamber.

Before installation, all nipples, fittings and devices must be clean and free of scale and burrs. Use pipe joint compound sparingly on male threads only.

OPERATION (Fig. A)

When the Accelerator is installed, it is connected to the dry pipe valve and to the system piping as shown in Figure C. When the Accelerator is set for service, the Valve Assembly is closed, the lower chamber is primed with water, the Accelerator Control Valve (4—Fig. C) is open, and the upper and lower chambers are pressurized with air at the same pressure as the system piping.

Operation of a sprinkler causes a pressure drop in the sprinkler system and in the lower chamber of the Accelerator. The pressure drop in the lower chamber is at a rate faster than can be communicated to the upper chamber through the orifice in the Restriction Chamber Assembly (6) thus creating a pressure differential between the upper and lower chambers.

The Diaphragm (29), reacting to the greater pressure in the upper chamber, pushes the Diaphragm Rod (30) causing the Weight (40) to pivot from its set position. As the Weight pivots, the Weight Lever (10) forces the Valve closure to open permitting air pressure in the system piping and in the intermediate chamber of the dry pipe valve to be equalized through the lower chamber of the Accelerator. The dry pipe valve then opens allowing water to flow to the open sprinkler. During operation of the system, the Anti-Flooding Device (2—Fig. C) will be closed to prevent debris in the system from entering the Accelerator.

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RESETTING

The Accelerator must be reset and the system restored to service as soon after operation as possible. The procedure is as follows:

1. Close the Accelerator Control Valve (4—Fig. C) and reset the dry pipe valve in accordance with instructions in Form TD19.

2. After the dry pipe valve has been reset, close the system's Main Control Valve and open the Main Drain Valve at the dry pipe valve.

3. Remove the Plugs (15, 14—Fig. C) from the Tee and Cross (10, 9—Fig. C), respectively. Remove the Air Chamber Plug (14—Fig. A) from the Accelerator.

4. Partially open the Accelerator Control Valve (4—Fig. C) to drain the Accelerator and connections. When water ceases to discharge, close the Accelerator Control Valve and replace the ¾ inch Plug (15—Fig. C) and the Air Chamber Plug (14—Fig. A). Do not replace the ½ inch Plug (14—Fig. C) previously removed from the Cross (9—Fig. C).

5. Remove the Anti-Flooding Device (2—Fig. C) from the Accelerator connections. Disassemble the device, thoroughly clean all interior parts, reassemble the unit and replace it in the connections.

6. Remove the Handhole Plate (3—Fig. A) from the Accelerator. Thoroughly clean the lower chamber and all interior components. Inspect the Valve Closure, Weight Lever Assembly and the Diaphragm Rod (30—Fig. A) for physical damage and freedom of movement. Be sure the ports in the Diaphragm Rod Guide (11—Fig. A) are clean and unobstructed.

7. Place the Valve Closure and Weight Lever Assembly in their respective set positions. Replace the Handhole Plate and Gasket (3, 38—Fig. A). Tighten the Hex Nuts (28—Fig. A) evenly.

8. Remove the Priming Plug (12—Fig. A) from the Handhole Plate. Slowly pour water into the priming cup in the Handhole Plate until water overflows.

9. Observe if water is leaking from the open outlet in the Cross (9—Fig. C) below the Accelerator. If there is no leaking, replace the Priming Plug in the Handhole Plate. However, if there are leaks, the cause must be remedied before proceeding; repeat the performance of Resetting Instructions 6—9 until there are no leaks.

10. Open the Accelerator Control Valve (4—Fig. C) to admit system air pressure to the Accelerator and again observe the open outlet of the Cross (9—Fig. C) for leaks. If there is no leaking, replace the ½ inch Plug (14—Fig. C) in the Cross. However, if there are leaks, close the Accelerator Control Valve and repeat the performance of Resetting Instructions 6—10 until there are no leaks.

    Caution: The priming water must be retained and the Accelerator Control Valve must be fully open before the Accelerator is placed in service.

11. When the Accelerator Air Pressure Gauge (3—Fig. C) and the air pressure gauge in the dry pipe valve trimmings indicate equal pressures, slowly open the system's Main Control Valve and close the Main Drain Valve as soon as water discharges.

The system is now set for service.

CARE AND MAINTENANCE

Principal concerns in the care and maintenance of the Accelerator is proper operation of the device and maintenance of the Diaphragm Plate, Valve and Restriction Chamber Assemblies in good operating condition.

Operation Test Procedure. Operation of the Accelerator can be tested in the following manner without operating the dry pipe valve.

1. Close the system's Main Control Valve and open the Main Drain Valve.

2. Remove the Plug (14—Fig. C) from the Cross (9—Fig. C).

3. Open the system's Inspector's Test Valve. Operation of the Accelerator should occur either within ten seconds or before a loss of 5 psi in the system air pressure.

4. Close the Inspector's Test Valve and the Accelerator Control Valve (4—Fig. C).

5. Reset the Accelerator in accordance with Resetting Instructions 3—11.

Operation of both the Accelerator and dry pipe valve can be tested by omitting performance of Operation Test Procedure Instruction 1 and 2. This will result in flooding the system thus requiring it to be drained before restoring it to service. *Such tests should not be conducted with systems exposed to freezing conditions.*

If, when tested, operation of the Accelerator did not conform to the limitations specified in Operation Test Procedure Instruction 3, check the Accelerator and trimmings for leaks and inspect the Anti-Flooding Device (2—Fig. C) to be sure it has not become obstructed. After verifying the integrity of the Accelerator, trimmings and the Anti-Flooding Device, the following inspection and test procedures apply.

Diaphragm Plate Assembly (Fig. A). Inspect the Diaphragm Plate Assembly in accordance with the following procedure:

1. Close the systems Main Control Valve and open the Main Drain Valve.

2. Close the Accelerator Control Valve (4—Fig. C) and remove the Air Chamber Plug (14) from Diaphragm Plate (2).

3. Remove the Diaphragm Plate (2) from the Accelerator. Unscrew the Diaphragm Rod Guide (11) from the back of the Plate.

4. Inspect the Diaphragm (29) to be sure it is flexible, has not become adhered to the Diaphragm Plate (2) and is free from physical damage or deterioration. Remove all rust, scale and other foreign material from interior surfaces of the Diaphragm Plate.

5. Inspect the Diaphragm Rod (30) and Diaphragm Rod Guide (11) to be sure there is no corrosion or other foreign matter that would restrict freedom of movement when the Accelerator operates.

6. Inspect the ports in the Diaphragm Plate (2) and Diaphragm Rod Guide (11). Clean the ports if they are clogged.

7. Replace parts disclosed by inspection to be defective. Using a new Diaphragm Washer (32), reassemble the Diaphragm (29), Diaphragm Rod (30) and Diaphragm Rod Guide (11) to the Diaphragm Plate (2). Engagement of the Diaphragm Rod Guide with the Diaphragm Plate must be achieved with a tightening torque of 50—100 foot—pounds.

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8. Remount the Diaphragm Plate Assembly to the Accelerator taking care to tighten the Hex Nuts (28) evenly. Proceed with inspection and testing the Valve Assembly. *Caution: The system's Main Control Valve should remain closed and the system's Main Drain Valve should remain open.*

Valve Assembly (Fig. A). Proper condition and operation of the Valve Assembly should be verified in accordance with the following procedure:

1. Remove the Handhole Plate (3) from the Accelerator. Remove all dirt, rust, scale, etc. that may have accumulated in the lower chamber and on the Valve Assembly.

2. Inspect all moving parts to be sure they are in good condition and that there are no obstructions to freedom of movement when the Accelerator operates. Replace any parts disclosed by inspection to be defective.

3. Remove the Plug (14—Fig. C) from the Cross (9—Fig. C) below the Accelerator and reset the Accelerator in accordance with Resetting Instruction 7—11.

4. Retest operation of the Accelerator by application of the Operation Test Procedure. If unsatisfactory tests result, the following instructions on the Restriction Chamber Assembly apply.

Restriction Chamber Assembly (Fig. A)
If, when retested, operation of the Accelerator continues to exceed the limitations specified in Instruction 3 of the Operation Test Procedure, it can be reasonably assumed that the Restriction Chamber Assembly (6) is not functioning properly and should be replaced. A new Restriction Chamber Cap Gasket (36) should be used when the Restriction Chamber Assembly is replaced.

Caution: It is not practical to clean, repair or replace components of the Restriction Chamber Assembly; attempts to do so are to be avoided.

ORDERING PROCEDURE

Orders for Model A—6 Accelerators, trimmings, accessories and replacement parts must reference the Product Symbol Numbers (PSN) for the items required. Product Symbol Numbers not identified in the ordering instructions are tabulated under "Product Symbol Numbers."

To Order Specify:
Grinnell Model A—6 Dry Pipe Valve Accelerator, PSN 52—003—1—001.

Grinnell Basic Trimmings for Model A—6 Accelerator with Model F302/F3021 Dry Pipe Valve, PSN 52—349—1—302.

Order Separately as Required:
Black ¾" pipe and cast iron fittings for completion of the connections between the Accelerator Control Valve (4—Fig. C) and the system piping.

Grinnell Extra Chamber No. (specify) for use with Model A—6 Accelerator, PSN (specify).

Grinnell (size) Extra Chamber Trimmings for Model A—6 Accelerator, PSN (specify).

The size of Extra Chamber Trimmings is determined by the size of the riser pipe to which the Extra Chamber is to be mounted. Extra Chamber Trimmings are available in 4, 5, 6 and 8 inch sizes.

If the Extra Chamber cannot be mounted to the system riser, order extra ¾" black pipe, cast iron fittings and supports separately as required for the connections between the Extra Chamber and the Accelerator.

PRODUCT SYMBOL NUMBERS

Following is a tabulation of Product Symbol Numbers of accessories and replacement parts for Model A—6 Accelerators and trimmings.

Accessories and Trimmings

Extra Chambers (Fig. B)

No. 2 Chamber . . . . . .52—053—1—002
No. 4 Chamber . . . . . .52—053—1—004
No. 6 Chamber . . . . . .52—053—1—006
No. 8 Chamber . . . . . .52—053—1—008

Extra Chamber Trimmings (Fig. D)

4"/100 mm . . . . . .52—349—1—313
5"/125 mm . . . . . .52—349—1—314
6"/150 mm . . . . . .52—349—1—315
8"/200 mm . . . . . .52—349—1—316

Replacement Parts

The Valve Assembly (w/Weight) listed in the following Accelerator replacement parts list consists of Items 8—10, 15—19 and 40—42 shown in Figure A. The Valve Assembly (w/o Weight) does not include Items 10, 15 and 40. The Weight Assembly consists of Items 10 and 40 shown in Figure A.

Accelerator (Fig. A)

2—Diaphragm Plate . . . .92—003—1—260
3—Handhole Plate . . . . .92—003—1—027
4—Restriction Chamber Cap . . . . . .92—003—1—025
6—Restriction Chamber Assembly . .92—003—1—044
8—Valve Cage . . . . . .92—003—1—006
9—Valve Body . . . . . .92—003—1—003
11—Diaphragm Rod Guide . . . . . .92—003—1—170
12—Priming Plug . . . . . .92—003—1—033
14—Air Chamber Plug . . .92—003—1—057
15—Weight Lever Screw . .92—003—1—041
16—Valve Washer . . . . . .92—003—1—104
18—Valve Pin . . . . . .92—003—1—023
29—Diaphragm . . . . . .92—003—1—100
30—Diaphragm Rod . . . .92—003—1—340
32—Diaphragm Washer . . .92—003—1—047
35—Valve Gasket . . . . . .92—003—1—016
36—Restriction Chamber Cap Gasket . . . . . .92—003—1—120
37—Diaphragm Plate Gasket . . . . . .92—003—1—014
38—Handhole Plate Gasket . . . . . .92—003—1—015
42—Valve Plug. . . . . .92—003—1—101
—Valve Assembly (w/Weight) . . . . . .92—003—1—045
—Valve Assembly (w/o Weight) . . . . . .92—003—1—046
—Weight Lever Assembly . . . . . .92—003—1—180

Items 5, 17, 19, 21, 23, 27, 28 and 41 listed in Figure A are common hardware items and should be obtained locally. Item 1 is not replaceable and thus is not included in the replacement parts list for the Accelerator. All other items listed in Figure A are available. To order replacement parts for the Accelerator, specify descriptions and Product Symbol Numbers, and indicate the model designation of the Accelerator for which the parts are required.

Basic Trimmings (Fig. C)

1—½" Check Valve . . . . .46—049—1—004
2—Anti-Flooding Device . .92—003—1—320
3—Accelerator Air Pressure Gauge. . . . . .92—343—1—010
4—¾" Globe Valve . . . . .46—047—1—005
8—Accelerator Support Plug. . . . . .92—003—1—328

Items listed in Figure C but not included in the preceding replacement parts list and all items listed in Figure D are common hardware items and should be obtained locally. Orders for replacement components for trimmings need only specify the descriptions and Product Symbol Numbers of the parts required.

USE THIS FORM WITH FORM TD19 FOR MODELS F302/F3021 DRY PIPE VALVES.

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